EECS 563: Introduction to Communications Networks

Fall 2026

What makes the Internet so difficult to design and operate?

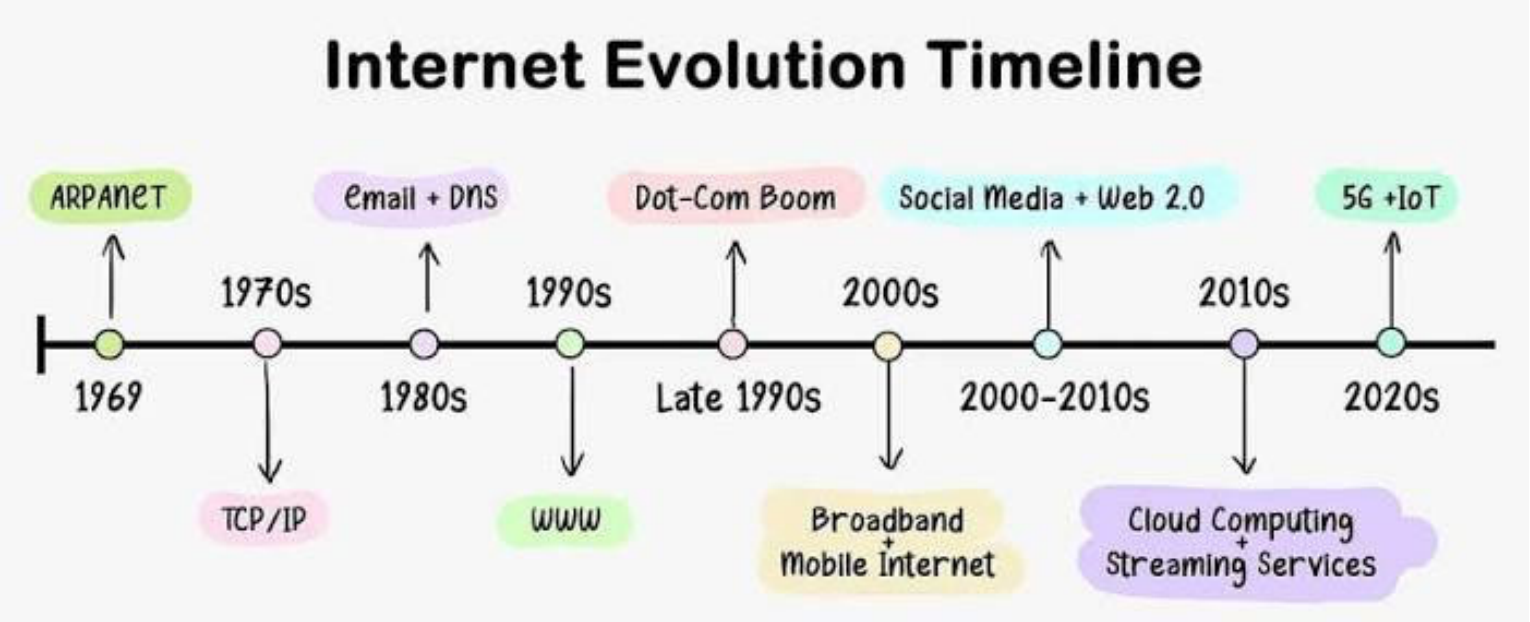
- If we had to *design the Internet from scratch*, what would make this difficult?

- How can billions of very different devices communicate efficiently and reliably across a global system that no single organization controls?

Potential Challenges

- Scale: Billions of devices and users
- Heterogeneity: Phones, laptops, IoT, servers; Wi-Fi, cellular, satellite, fiber
- Shared resources: Millions of flows compete for limited bandwidth
- Dynamics & failures: Traffic changes, users move, links and devices fail
- Different requirements: Web, video, gaming, AI, autonomous systems all need different performance
- Decentralization: The Internet is made of many independently operated networks

Despite these challenges…

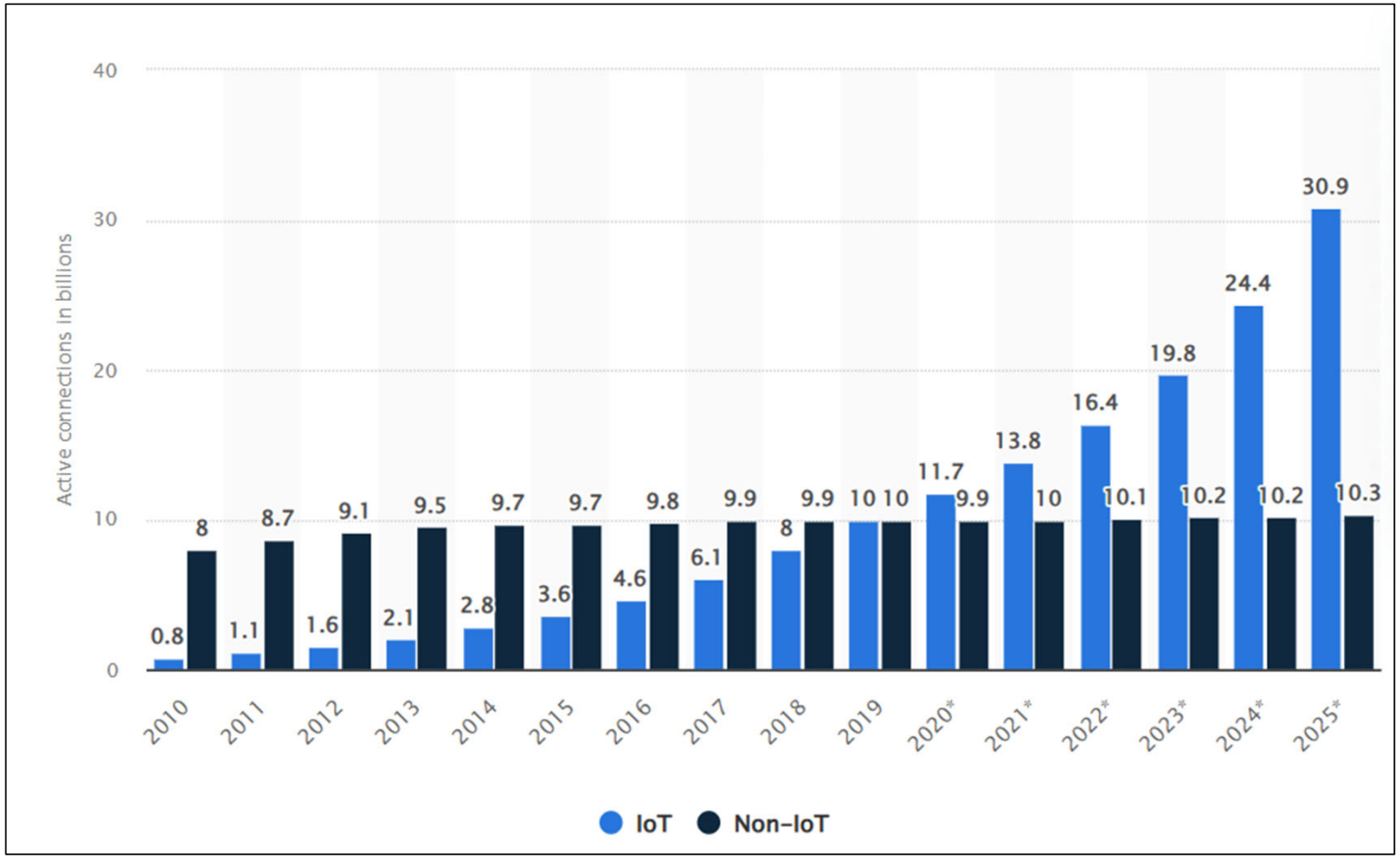


Why do we need to learn about modern communications networks?

Question

- Approximately how many devices (IoT, cellular, etc.) were connected to the Internet in 2025?
  A. 500 million
  B. 1 billion
  C. 10 billion
  D. Over 25 billion

Number of connected devices



Source: Statista

Number of connected devices

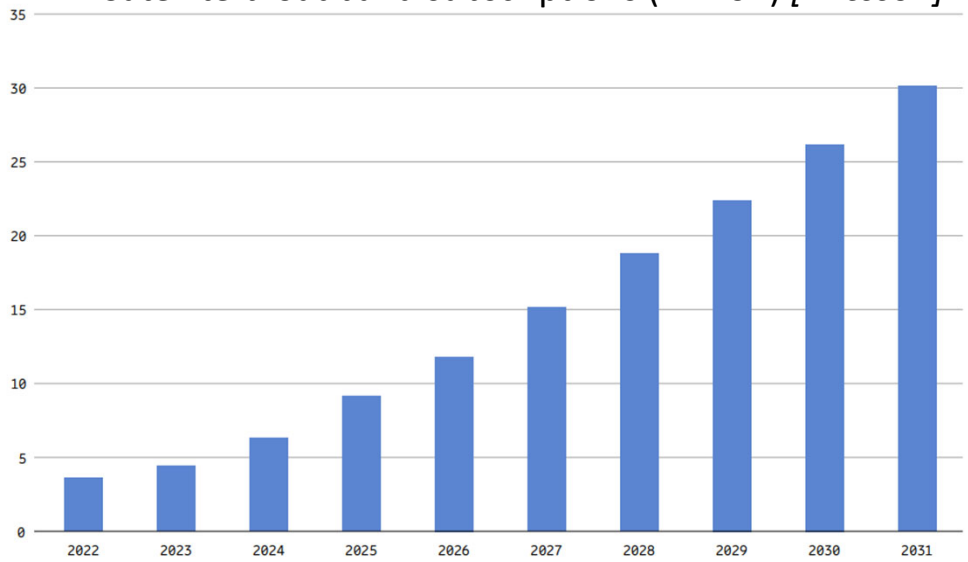
- IoT connections
  - The total number of Internet of Things (IoT) connected devices: More than **30.9 billion units by 2025**
  - *Examples of IoT:* connected cars, smart home devices, and connected industrial equipment.
  - Rollout of 5G and 6G accelerates the pace with which IoT deployment occurs

- Non-IoT Connections
  - Smartphones, laptops, and computers,
  - Just over **10 billion units in 2025**
  - The proportion of mobile subscriptions that are 5G increased from one-quarter at the end of 2024 to one-third in 2025!

Satellite Broadband Subscription

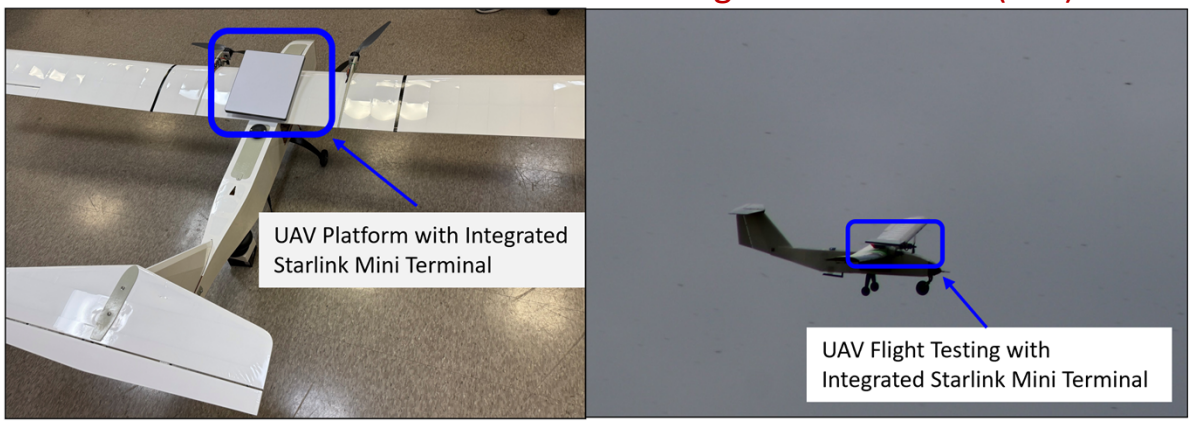
- In recent years, there has been a growth in satellite networks, in particular large constellations of low-Earth orbit (LEO) satellites.
- These networks can provide multiple services such as direct-to-device connectivity and fixed broadband internet.

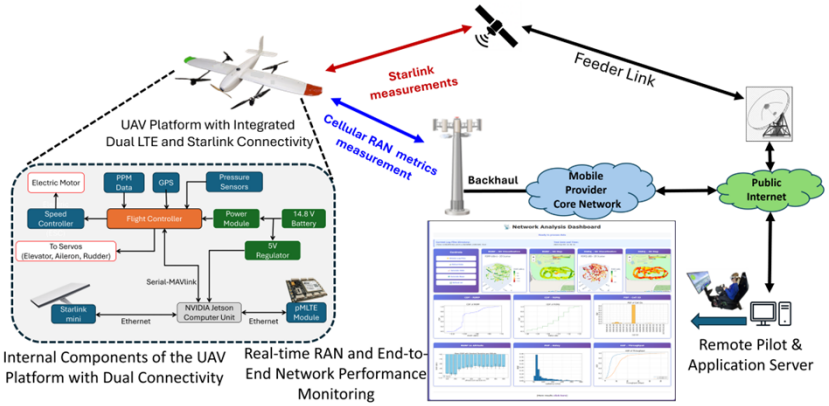
Satellite broadband subscriptions (million) *[Ericsson]*

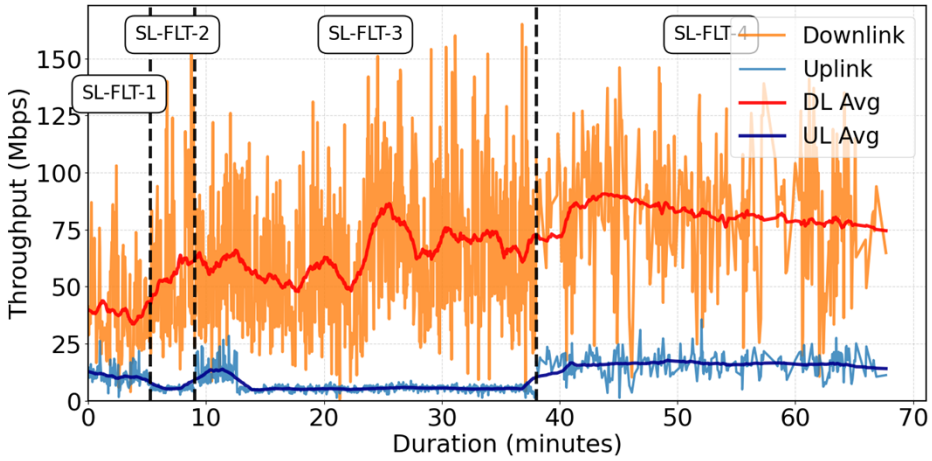


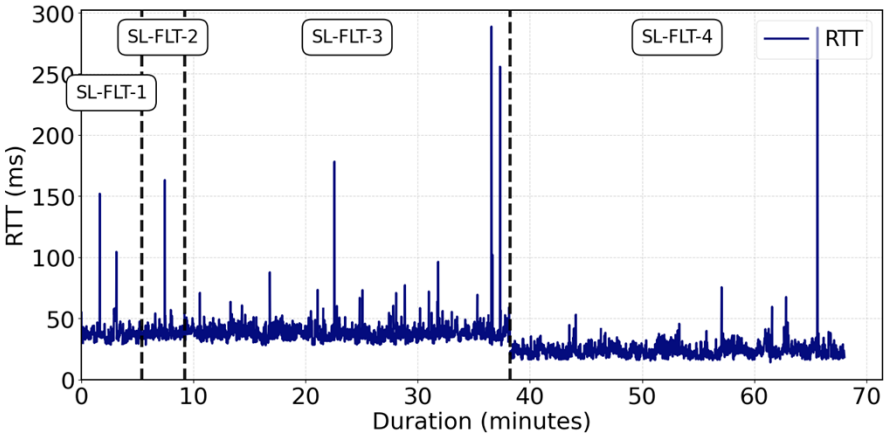
Satellite Connections for New Use-cases

Pictures from our collaborations with KU Flight Research Lab (FRL)









How Much Data Flows Through the Internet?

Data-Intensive Applications

- Internet infrastructure keeps expanding.
- Total monthly global mobile network data traffic reached 188 EB.
  - 1 EB = 10^{18} bytes
  - 1 EB = 1 Billion GB
  - 1 EB = 1 million TB



Data-Intensive Applications

- **Question:** What generates the most Internet traffic worldwide?
  A. Video traffic
  B. Web browsing
  C. File sharing
  D. Email

➢ At the end of 2025, video traffic accounted for 76 percent of all mobile data traffic.

What New Applications Are Shaping the Internet?

AI Changes Internet Usage

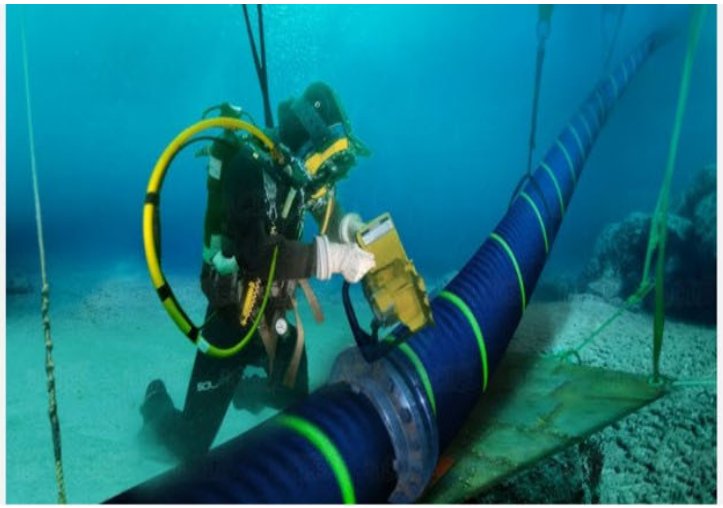
- Getting close to a future where "Googling things" is no longer the primary way of getting information
- Generative AI replacing search in many tasks
- Explosive growth in 2025: usage was doubled across businesses for everyday search.
- AI assistants becoming standard interface
- Summarize information, generate answers, or complete tasks without ever clicking through to a webpage.

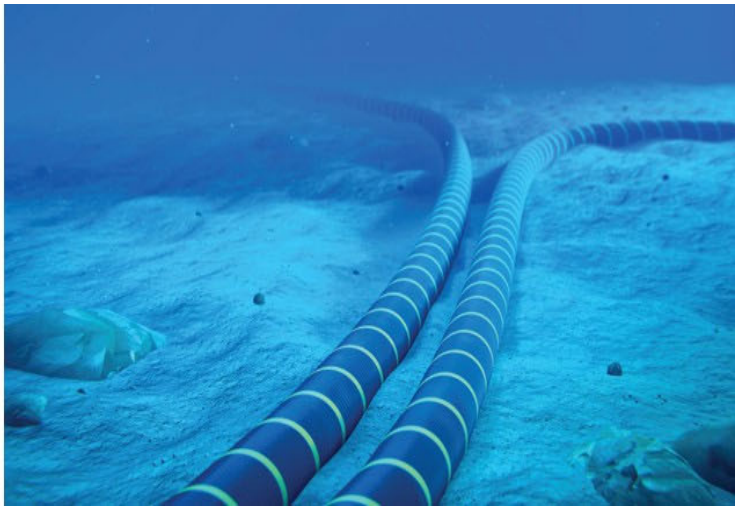
Question

- Why is networking essential for Gen-AI models like ChatGPT?
  A. To process user data locally
  B. To ensure fast data transfer between servers and users
  C. To replace traditional CPUs with GPUs
  D. To reduce Internet usage

Internet and Physical Infrastructure

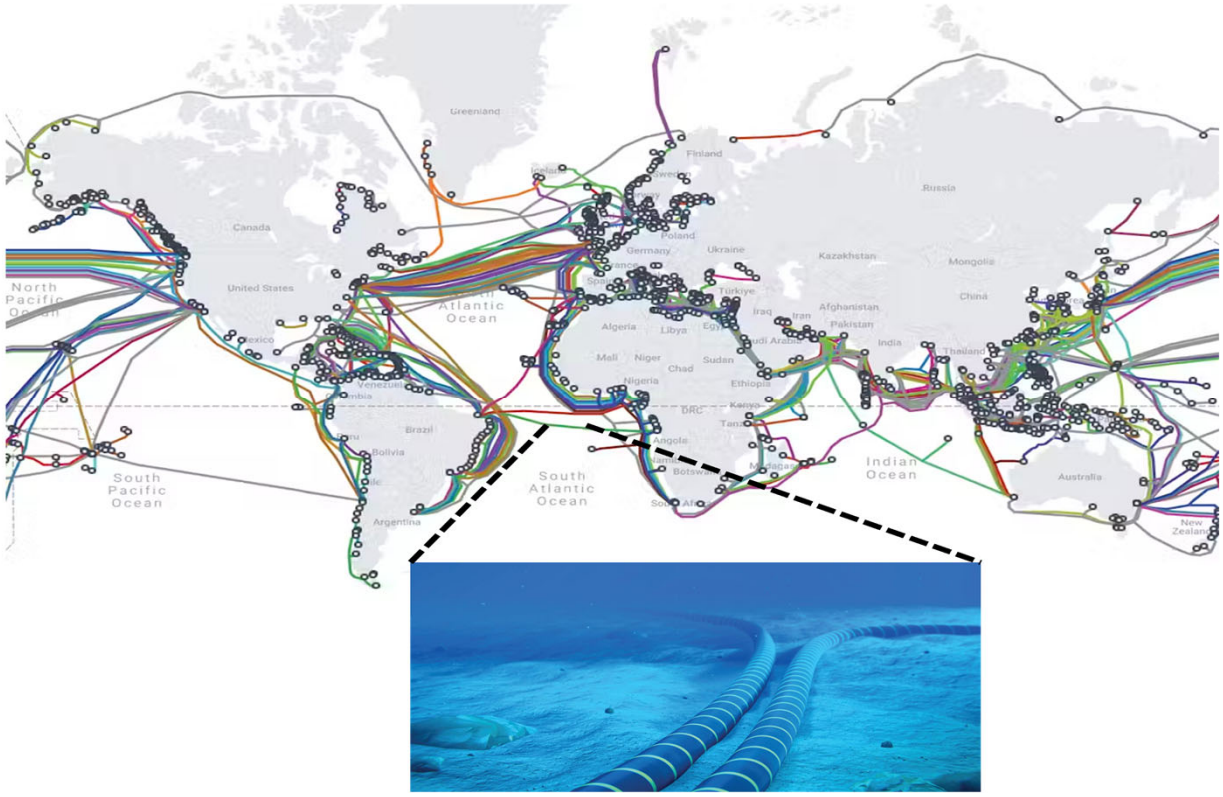
Backbone of the Internet: Submarine Cables





Physical geography matters!

Backbone of the Internet: Submarine Cables



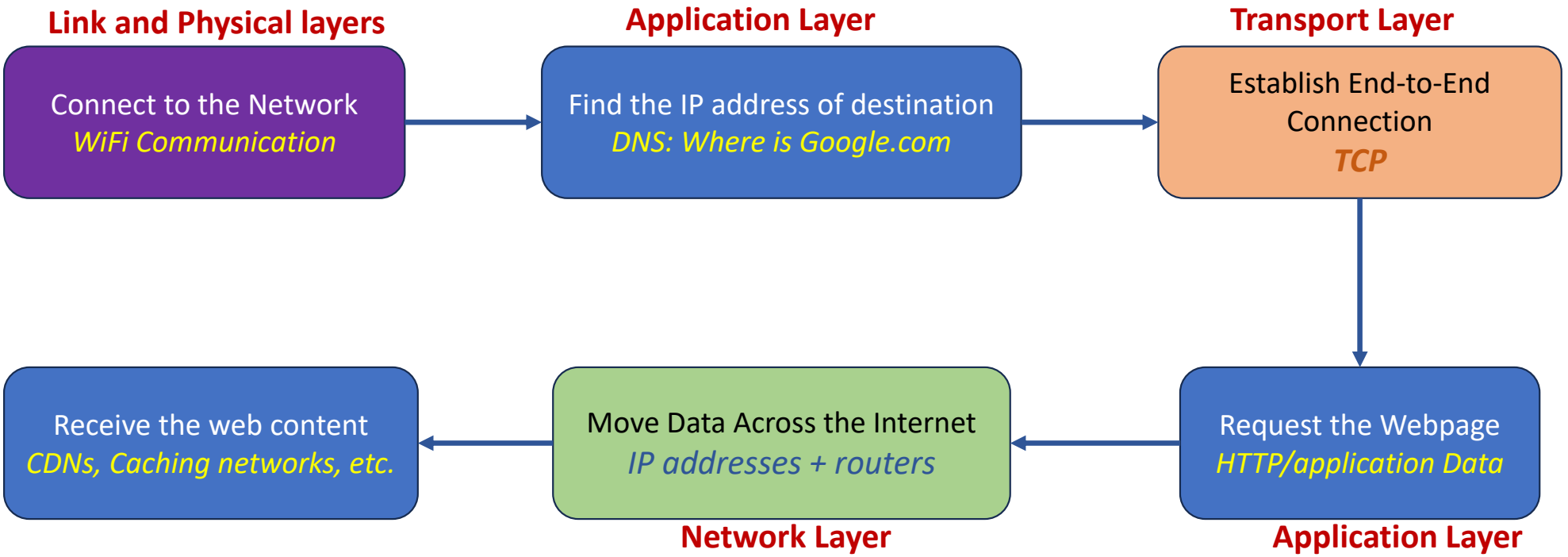
What are we going to learn in this class?

What happens when we connect to *www.Google.com*?

- Our laptop/phone➔??? ➔ KU WiFi (or LTE/5G) ➔ ??? ➔ www.Google.com

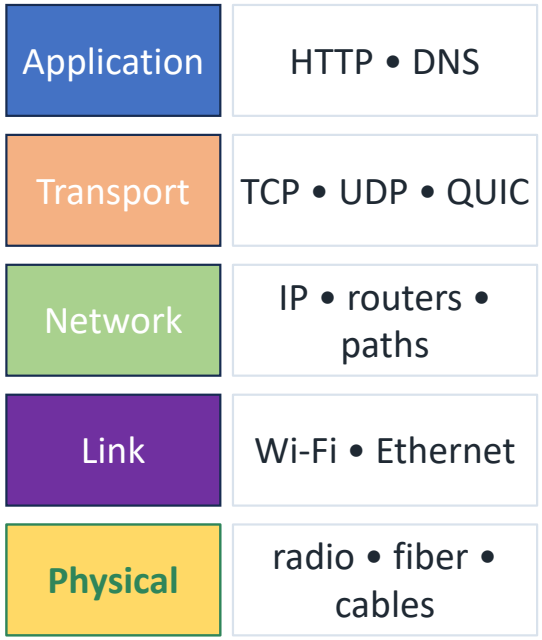
- We connect to Wi-Fi ➔ Open browser ➔ Search in www.Google.com
- What is happening before we can receive any data from the web server?
- *A lot happens behind the scene!*

What happens when we connect to Google.com?



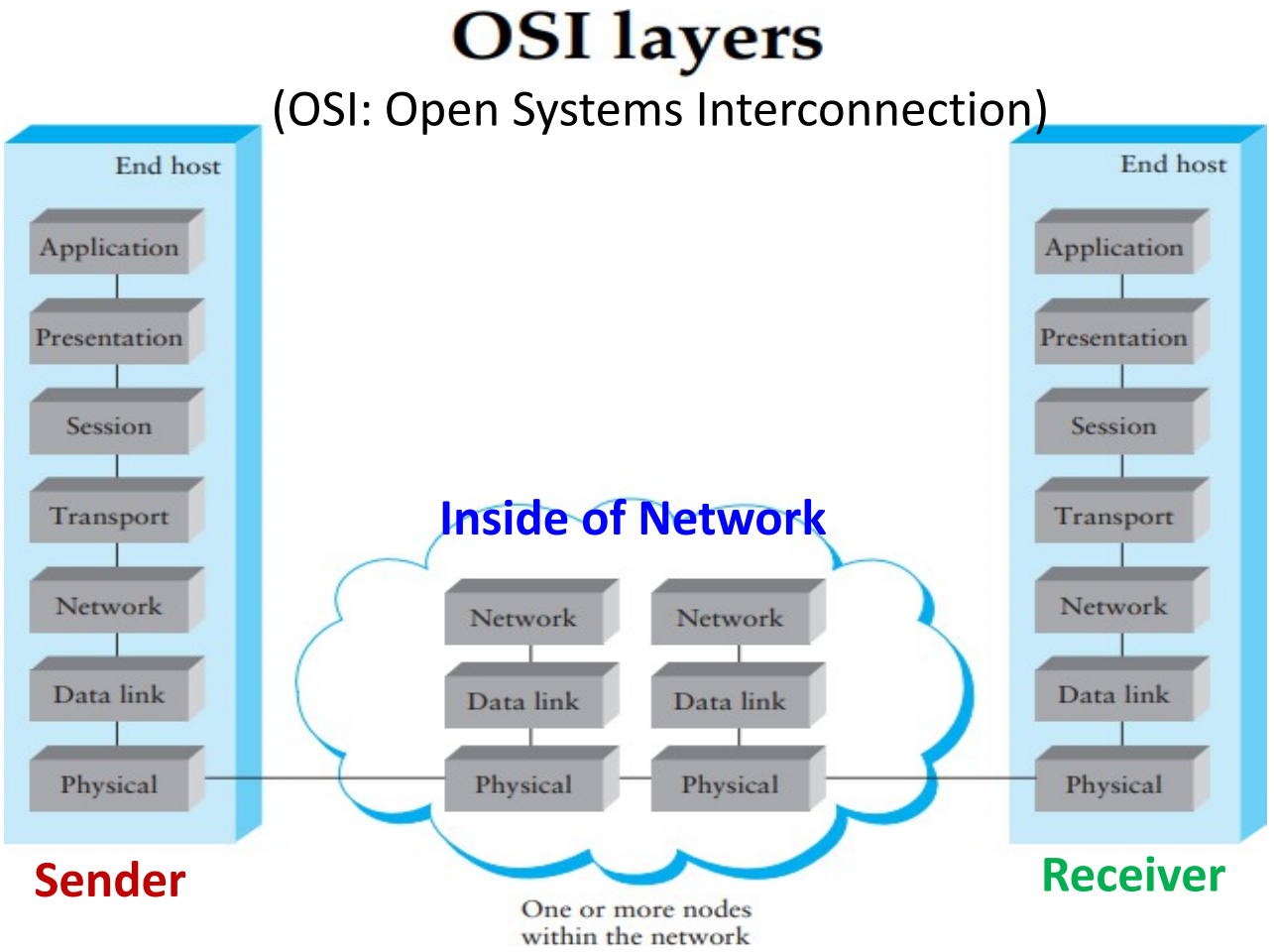
What happens when we connect to Google.com?

Five-layer Internet model

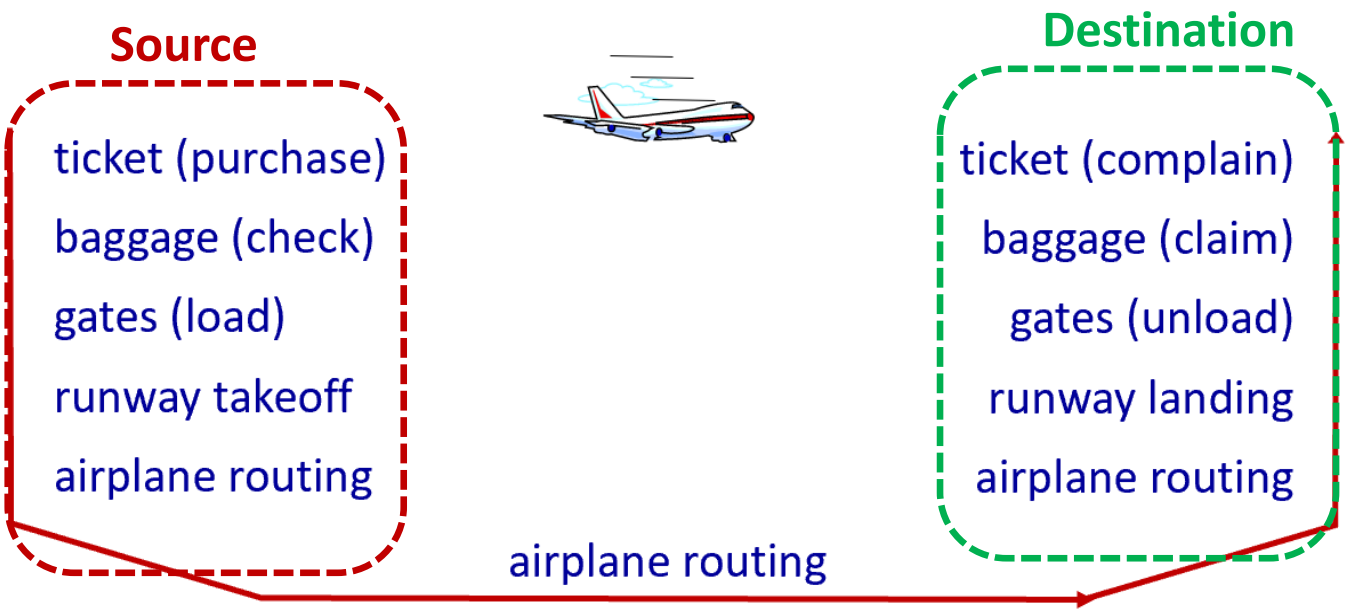


By the end of the semester, we will be able to explain the full path:
Connecting a new laptop to the Internet ➔ Receive content from the Internet

A Layered Architecture



Analogy: Air Travel System



airline travel: a series of steps, involving many services

Analogy: Postal Service

- Bob (the Application) puts a letter in an envelope
- Bob writes Larry's address on the outside and puts it in his mailbox.
- The mailman (the Transport) picks it up and takes it to the Post Office (the Network).
- Here, they look at the address (the Data Link layer) and decide how to send it (the Physical transportation means).
- At the destination, the process proceeds in the reverse order.
- They unload it from the train, look at the address, decide which of many mailmen to give it to, and take it to Larry's, who opens the letter, and reads the message.

Analogy: Postal Service



Visualization generated by GenAI

EECS 563

- Basics of data communications, network architecture, and internetworking
- Layered architecture of network protocols, focusing on the five-layer Internet protocol stack.
  1. Application Layer: like HTTP, SMTP, POP3, IMAP, and DNS, and their role in enabling internet services.
  2. Transport layer concepts, services, and TCP/UDP
  3. Network layer control and protocols
  4. Link layer control and protocols
  5. Physical layer: Wireless communication principles and technologies

- Designing basic and advanced communication protocols

Course Outline

- Module 1: Introduction
- Module 2: The Application Layer
- Module 3: The Transport Layer
- Module 4: The Network Layer
- Module 5: The Link Layer and Local Area Networks
- Module 6: Physical layer; Wireless communications

Takeaway Message

- Supporting billions of heterogeneous devices ➔ Scalability & interoperability
- Huge amounts of data traffic ➔ Capacity & Congestion
- Supporting new applications ➔ Different performance requirements
- Changing conditions and failures ➔ Reliability
- Global and decentralized Infrastructure ➔ Routing & Coordination